

Do Unknown Scenarios Arrive as a Poisson Process?

A Layered-Scenario Simulation Framework for the Statistical Foundations of Automated-Vehicle Validation*

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Abstract

Safety arguments for automated driving systems commonly assume that manifestations of functional insufficiencies—hazardous behaviours triggered by scenarios the system cannot handle—occur according to a Poisson process. The assumption underpins two load-bearing calculations: the validation mileage required to demonstrate an acceptable residual risk, and the decomposition of vehicle-level acceptance criteria into per-scenario-class targets. Verifying it empirically requires far more independent observations of rare events than any physical fleet can collect, so we verify it in simulation. We present an event-driven, fully reproducible simulator built on a six-layer scenario model: layers evolve as independent semi-Markov processes over generated element catalogues with three rarity classes, per-element Gamma sojourn distributions, and exactly configured class selection masses. Unknown scenarios are modelled in two ways: as rare per-transition selections of unknown elements, and as never-anticipated exact combinations of three to six known rare elements. We prove, via renewal thinning and superposition arguments, that episode counts converge to a Poisson process as the triggering probability vanishes, with a dispersion deviation of only 0.2–0.3% at realistic parameters, and we confirm the prediction empirically: replication, window-count, and inter-arrival tests all retain the Poisson hypothesis for the element-route mechanism. For combination-type unknowns we identify a fine-scale clustering effect—rapid re-openings of the same rule inside persistent slow-layer states—that inflates the dispersion index to 1.36; counting distinct manifestations instead of raw episodes restores full Poisson compatibility. A 21-study, 1,160-simulation sensitivity analysis ranks the model factors that drive unknown-scenario exposure. We discuss the practical boundary of the assumption: it holds conditional on a fixed operational world, whereas across-world uncertainty makes counts mixed-Poisson and overdispersed, which validation-mileage budgets should accommodate.

Keywords: automated vehicles; safety validation; scenario-based testing; unknown scenarios; Poisson process; overdispersion; renewal theory; simulation.

1 Introduction

Demonstrating that an automated driving system (ADS) is acceptably safe is, at its core, a statistical problem. Field data indicate that safety-relevant events are rare; consequently, purely mileage-based demonstration of reliability requires driving distances that are infeasible in practice—hundreds of millions to billions of miles for direct statistical proof of a fatality-rate improvement [1, 2]. Industrial validation programmes therefore combine scenario-based testing [3, 4], functional-safety and safety-of-the-intended-functionality (SOTIF) processes [5, 6], and statistical extrapolation from limited fleets [7].

Within such programmes, a specific probabilistic assumption is routinely made: that manifestations of *functional insufficiencies* (FIs)—behaviours in which the system fails to cope with the situation it

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encounters—occur according to a homogeneous Poisson process over driven distance. At an industrial developer of automated driving systems whose validation methodology motivated this work, the assumption enters the safety case at two points. First, it determines the *validation mileage*: under a Poisson model, the number of FI manifestations k observed in d driven miles is $k \sim \text{Poisson}(\lambda d)$, and standard confidence arguments convert an observed count into an upper bound on the rate λ , hence into a required demonstration mileage. Second, it justifies the *decomposition of acceptance criteria*: a vehicle-level tolerable rate can be split additively across scenario classes only if the underlying counting processes superpose the way Poisson processes do.

The assumption is attractive because it is simple and because the rare-event heuristic (“rare events are Poisson”) is folklore. It is also consequential when wrong. If counts are *overdispersed*—variance exceeding the mean—occurrences cluster, observed event-free mileage overstates safety, and validation targets computed from Poisson statistics are optimistic. If counts are *underdispersed*, targets are conservative and resources are wasted. Distinguishing these regimes empirically requires estimating both the mean and the variance of counts across many independent mileage windows. With unknown scenarios occurring at rates well below one per thousand miles, no test fleet provides the required replication; the verification can, at present, only be carried out in simulation.

This paper undertakes that verification. Our object of study is the occurrence process of *unknown scenarios*: scenarios outside the design-time catalogue, the primary trigger of FIs in the SOTIF sense (“unknown–unsafe” conditions) [6]. We make four contributions.

(i) *A purpose-built simulation framework.* We formalize a layered scenario model—six independent semi-Markov layers over generated element catalogues with three rarity classes (*common*, *rare*, *unknown*)—with exactly configured class selection masses, one Gamma sojourn law per element, and an event-driven engine that simulates 20,000 hours (one million miles) of operation in about twelve seconds of wall-clock time, bit-identically reproducibly.

(ii) *Two mechanisms for unknown-scenario generation.* We model unknowns first as rare per-transition selections of unknown-class elements (the *element route*), and second as *exact rare combinations*: selected sets of three to six known rare elements from distinct layers whose joint occurrence was never anticipated. The second mechanism operationalizes the observation that unknown scenarios frequently consist of individually known conditions in an unforeseen constellation.

(iii) *Theory and verification.* Using renewal thinning [8, 9] and superposition [10] arguments we prove that element-route episode counts converge to a Poisson process in the rare-event limit and quantify the finite-probability deviation (0.2–0.3% in the dispersion index at our parameters). A three-part statistical battery—replicated runs, window counts, and inter-arrival distances—retains the Poisson hypothesis on all tests. For the combination mechanism we identify, explain, and repair a fine-scale clustering phenomenon, showing that Poisson behaviour holds at the level of distinct manifestations.

(iv) *Sensitivity analysis.* A suite of 21 one-factor studies comprising 1,160 simulations ranks every model control by its effect on unknown-scenario exposure, separating factors that matter (class percentages, combination budgets, duration scales) from exact invariances predicted by the theory.

The paper is organized as follows. Section 2 reviews the validation context and the layered scenario model. Section 3 specifies the simulation framework. Section 4 develops the theoretical analysis. Section 5 describes the experimental design, Section 6 reports results, and Section 7 discusses implications, limitations, and the practical boundary of the Poisson assumption. Section 8 concludes.

2 Background and related work

2.1 Scenario-based validation and the mileage problem

Scenario-based approaches decompose the open driving task into scenarios: temporally evolving arrangements of road layout, environment conditions, and the behaviour of the ego vehicle and other road users [4]. Validation then argues coverage over a scenario space rather than over raw mileage [3, 11]. The approach was systematized by, among others, the German PEGASUS project, and is now embedded

in the SOTIF standard’s goal of shrinking the “unknown–unsafe” region of the scenario space [6].

Statistical demonstration nevertheless remains necessary, because scenario catalogues are finite while the operational world is not. Kalra and Paddock [1] showed that demonstrating, with 95% confidence, a fatality rate better than the human benchmark would require on the order of 275 million failure-free miles, and correspondingly more to bound rarer events; Wachenfeld and Winner [2] reach similar conclusions for a motorway chauffeur. These calculations *assume* Poisson statistics: their confidence bounds are exponential/Poisson quantiles. The assumption itself, to our knowledge, has not previously been stress-tested against a generative model of how unknown scenarios actually arise from the structure of driving.

2.2 The layered scenario model and unknown scenarios

A widely adopted way to structure the scenario space is the layered model, in which a driving scene is described as the combination of one active state in each of several quasi-independent layers—road network, temporary modifications (e.g. work zones), the ego manoeuvre, surrounding road-user behaviour, environmental conditions, and additional triggering conditions [4, 12]. The scene at any instant is an element of the Cartesian product of the layers; driving is a chain of such scenes [11].

The layered view also organizes thinking about *where unknown scenarios come from*. Three routes are distinguishable: an unknown element inside a known layer (a phenomenon the taxonomy lacks); an entirely unknown layer (a missing descriptive dimension); and an *unknown combination of known elements*—each constituent is catalogued, but their joint occurrence was never anticipated or tested. The third route is arguably the most insidious for mature programmes, whose per-layer catalogues are extensive but whose combinatorial coverage is necessarily sparse: with even a few dozen elements per layer, the number of C_k cross-layer combinations grows into the millions, only a vanishing fraction of which any test campaign visits. Both routes 1 and 3 are modelled in this paper.

2.3 The Poisson assumption and its consequences

If unknown-scenario occurrences form a homogeneous Poisson process of rate λ per mile, three operationally useful properties follow: counts in fixed windows are Poisson distributed with variance equal to the mean; inter-occurrence distances are exponential, hence memoryless; and independent sources superpose to a Poisson process whose rate is the sum of the parts. The first property supports mileage-based confidence bounds; the second underlies “time between events” monitoring; the third legitimizes additive decomposition of acceptance criteria.

The natural diagnostic is the *index of dispersion* $D = \text{Var}(k)/\text{E}(k)$ estimated over disjoint windows, with $D \approx 1$ under the Poisson hypothesis, $D \gg 1$ indicating risky clustering and $D \ll 1$ indicating conservatism [13]. A useful reference point for what genuine violation looks like: in an early version of our simulator that counted every new element *tuple* during an unknown period as a separate “encounter”, window counts exhibited $D \approx 12$ —an order-of-magnitude clustering artefact produced purely by the counting convention (Section 3.5).

3 The simulation framework

3.1 Design goals and overview

The framework was designed around four requirements. *Structural fidelity*: the generative process should reflect the layered scenario model rather than postulate the answer (nothing in the engine samples from a Poisson law). *Controlled rarity*: the prevalence of common, rare, and unknown phenomena must be configurable exactly, so that verification targets are unambiguous. *Scale*: testing dispersion requires hundreds of independent windows, hence millions of simulated miles. *Reproducibility*: every random draw must be attributable to a seeded stream so that results are bit-identical across machines and across checkpoint boundaries.

Table 1: Layer configuration and realized element composition in the reference run. Counts are drawn per layer from the configured range; rarity classes are allocated from the configured 70/20/10% split by the largest-remainder method. Baseline sojourn parameters give each layer’s characteristic time scale before per-element modulation (Section 3.3).

Layer	Count range	Built	Common/rare/unknown	Baseline mean (s)	Baseline var (s ²)
street	12	12	9/2/1	300	20,000
temporal modifications	40–45	45	32/9/4	600	90,000
ego manoeuvre	20–24	20	14/4/2	30	400
RU manoeuvre	20–24	22	16/4/2	45	900
environmental conditions	22	22	16/4/2	1,800	500,000
triggering conditions	80–90	83	58/17/8	120	8,000
Total		204	145/40/19		

A run proceeds in two phases: *world construction* builds the layers, element catalogues, transition vectors, sojourn distributions, and—for the combination mechanism—the set of planted unknown-combination rules; the *event loop* then advances the six-layer state through simulated time, opening and closing unknown-scenario episodes, until a configured horizon (20,000 h in the reference configuration) is reached exactly.

3.2 World construction

Layers and elements. The model instantiates the six layers in Table 1: street, temporal modifications, ego manoeuvre, road-user (RU) manoeuvre, environmental conditions, and triggering conditions. Each layer k receives a generated element catalogue E_k whose size is drawn (seeded) from a configured integer range; elements carry stable synthetic identifiers (street_000, ego_003, trigger_042, ...). We deliberately use abstract catalogues rather than semantic ones: the statistical questions studied here depend on the *structure* of the process (layer counts, rarity masses, sojourn scales), not on element labels, and abstract catalogues let every structural parameter be swept freely in the sensitivity studies.

Rarity classes. Every element belongs to exactly one of three classes—*common*, *rare*, or *unknown*—controlled by two user-configured percentage maps that each sum to 100%. The *element-class percentages* (reference value 70/20/10) determine how many catalogue entries belong to each class, with integer counts allocated by the largest-remainder method; across the 204 elements of the reference build the realized shares are 71.1/19.6/9.3%. The *selection-class percentages* (also 70/20/10 in the reference configuration) determine the total transition probability mass of each class, exactly.

Transition vectors. Each layer holds a permanent probability vector $q^{(k)}$ over its elements. When a sojourn expires, the next element is drawn from $q^{(k)}$, independently of all other layers; self-transitions are permitted. To make elements unequal without disturbing class totals, $q^{(k)}$ is sampled once at build time from a Dirichlet distribution with concentration $c\pi$ (reference $c = 20,000$, π the normalized target masses) and then rescaled class-by-class so that each class’s total mass equals its configured selection percentage *exactly*. The empirical selection share of unknown-class elements across the reference run is 10.02% against the 10% design target; the permanent class masses are exact by construction.

3.3 Sojourn durations

Every element i carries its own Gamma sojourn distribution. Its mean is placed inside a non-overlapping band determined by its rarity class,

$$\mu_i = \mu_k \cdot m_{r(i)} \cdot (1 + su_i), \quad \sigma_i = \text{CV}_{r(i)} \mu_i, \quad (1)$$

Table 2: Rarity-class duration profiles (reference configuration).

Class	Mean multiplier m_r	Coefficient of variation CV_r	Within-class spread
common	1.50	0.45	$\pm 10\%$
rare	0.75	0.60	$\pm 10\%$
unknown	0.25	0.80	$\pm 10\%$

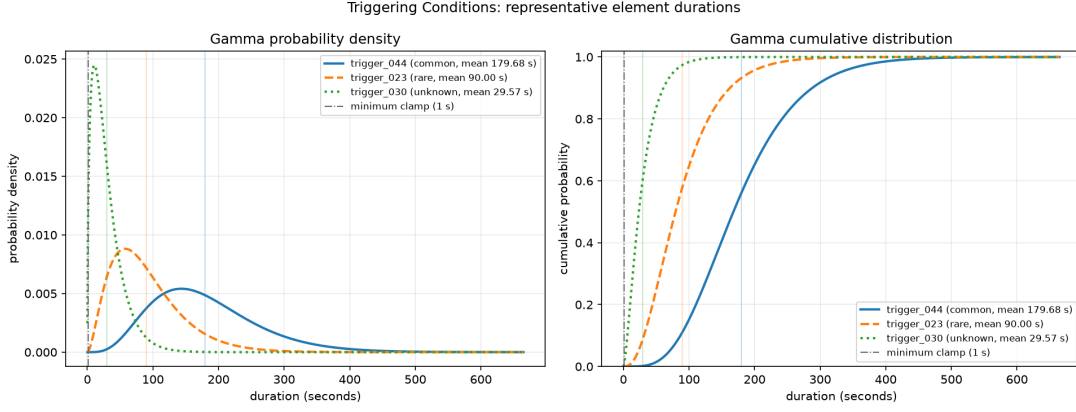


Figure 1: Per-element Gamma sojourn laws for representative elements of the triggering-conditions layer (baseline mean 120 s): probability density (left) and cumulative distribution (right) for a common (179.7 s), rare (90.0 s), and unknown (29.6 s) element, with the 1 s minimum clamp.

where μ_k is the layer baseline of Table 1, m_r and CV_r are the class multiplier and coefficient of variation of Table 2, $s = 0.10$ is the within-class spread, and $u_i \in [-1, 1]$ is a deterministic position derived from catalogue order. Gamma shape and scale follow as $\kappa_i = \mu_i^2 / \sigma_i^2$ and $\theta_i = \sigma_i^2 / \mu_i$; all draws are clamped below at 1 s. Within every layer the bands are disjoint: each common mean exceeds each rare mean, which exceeds each unknown mean—unknown states are brief by design. Figure 1 shows the resulting laws for representative elements of the triggering-conditions layer. Class coefficients of variation between 0.45 and 0.80 give Gamma shapes $\kappa \approx 1.6$ –4.9, i.e. sojourns more regular than exponential, which matters for the finite- p analysis of Section 4.

3.4 Unknown-scenario mechanisms

Mechanism A: the element route. In the first mechanism, unknown scenarios arise when a layer selects one of its unknown-class elements. The configuration used for the formal verification battery assigns each unknown-bearing layer a small unknown transition mass of 0.4% (realized 0.38–0.44% across layers), so that each layer transition independently opens an unknown episode with probability $p \approx 0.004$. This mechanism realizes route 1 of Section 2 and matches the independence structure assumed by the theory of Section 4 exactly.

Mechanism B: exact rare combinations. The second mechanism, the default of the current framework (version 6), realizes route 3: unknowns are *never-anticipated constellations of individually known elements*. At build time the simulator plants R combination rules—in the reference configuration 40 of size 3, 30 of size 4, 20 of size 5, and 10 of size 6—each consisting of one *rare* element from each of k distinct layers and always including a rare triggering-condition element. Rules are drawn uniformly without replacement by sampling candidate ordinals and unranking them, so the combinatorially large candidate space is never materialized. The stationary mass of a rule (the product of its members’ selection probabilities) ranges from about 10^{-5} for size-3 rules down to about 10^{-10} for size-6 rules.

A rule *matches* the current scene if and only if the set of active rare-class elements equals the rule’s element set exactly, and no unknown-class element is active in any layer (all remaining layers

are therefore in common states). Exactness has two consequences: a size-3 rule embedded in a richer rare state does not fire, and every episode is attributable to exactly one rule. Under this mechanism, unknown-*class* elements never open episodes themselves; their only role is to disqualify matches while active.

3.5 Episode semantics

The framework counts *episodes*: maximal continuous periods during which the unknown condition persists—for mechanism A, the period an unknown element remains active; for mechanism B, the period the active rare set stays equal to one rule. Layer self-transitions that leave the condition intact continue the episode rather than re-triggering it. Each episode records its opening and closing time and mileage.

This convention is not innocuous. An early version counted every new cross-layer element tuple observed during an unknown period as a separate “encounter”; because fast layers churn while a slow unknown persists, one real condition then spawned bursts of counts, and window counts showed a dispersion index near 12. Redefining the unit of counting as the episode is what makes the occurrence process statistically well-posed: opens become rare, short, well-separated events—the natural analogue of “one FI manifestation”. Section 6.4 shows that a residue of the same phenomenon reappears, in much weaker form, for mechanism B, and how it is resolved.

3.6 Event-driven execution and reproducibility

The engine advances the world from layer expiry to layer expiry rather than in fixed time steps. Each iteration (i) finds the earliest remaining sojourn $\Delta t = \min_k \tau_k$, clipped at the simulation horizon; (ii) advances time and mileage (at a constant configured 50 mph, mileage is derived from time and is not the stopping criterion); (iii) redraws, for each expired layer, the next element from $q^{(k)}$ and a fresh Gamma sojourn; and (iv) re-evaluates the unknown-scenario condition on the new scene, opening or closing episodes as required. The reference run covers 20,000 h (72×10^6 s; one million miles) in 4,079,575 events—one event per ≈ 18 simulated seconds—in 11.7 s of wall-clock time.

Randomness is split across seven independently seeded streams (element counts, rarity placement, Dirichlet sampling, durations, initial state, transition sampling, and rule selection), so any configuration reproduces bit-identical results; checkpoint/resume execution is bit-identical to an uninterrupted run. The implementation ships with 39 automated tests covering configuration validation, exact-set matching, triggering-condition inclusion, unknown-element disqualification, exact time stopping, and checkpoint equivalence.

4 Theoretical analysis

We first establish what the Poisson hypothesis *should* mean for this model, conditional on a built world (fixed catalogues, vectors, and sojourn laws). Proofs are elementary renewal theory; we sketch them for completeness. Throughout, a layer has i.i.d. Gamma sojourns X with mean μ , shape κ , and scale θ ($\kappa\theta = \mu$), and at each renewal the next element is drawn from the fixed vector q independently of the past.

Proposition 1 (Single layer: thinned renewal limit). *Let episodes open at those transitions of the layer’s renewal process that select an unknown element not already active, which happens with probability $p = m - s$ per transition, where $m = \sum_{u \in U} q_u$ is the unknown mass and $s = \sum_{u \in U} q_u^2$ removes self-hops. Then, as $p \rightarrow 0$ with the sojourn law fixed, the process of episode openings, rescaled by its rate $r = p/\mu$, converges to a homogeneous Poisson process of unit rate.*

Proof sketch. The waiting distance between consecutive openings is a geometric sum $W = X_1 + \dots + X_G$ with $G \sim \text{Geometric}(p)$, so the openings form a renewal process with

$$\mathbb{E}[e^{-\lambda W}] = \frac{p \varphi(\lambda)}{1 - (1-p) \varphi(\lambda)}, \quad \varphi(\lambda) = (1 + \theta \lambda)^{-\kappa}.$$

226 Evaluating at $\lambda p/\mu$ and expanding $\varphi(\lambda p/\mu) = 1 - \lambda p + O(p^2)$ (using $\kappa\theta = \mu$) gives $E[e^{-\lambda pW/\mu}] \rightarrow$
 227 $(1 + \lambda)^{-1}$, the Laplace transform of $\text{Exp}(1)$. A renewal process with exponential inter-arrivals is Poisson;
 228 this is the classical thinning/rarefaction theorem of Rényi [8] (see also 9). Consecutive opening indicators
 229 are 1-dependent (adjacent transitions share one element); the Chen–Stein method [14] bounds the induced
 230 total-variation error by $O(p)$, of the same negligible order. $\square$

231 **Proposition 2** (Superposition across layers). *Let the K unknown-bearing layers be independent, layer*
 232 *k opening episodes at asymptotic rate $r_k = (m_k - s_k)/\mu_k$. As all $p_k \rightarrow 0$, the merged opening process*
 233 *converges to a Poisson process of rate $\Lambda = \sum_k (m_k - s_k)/\mu_k$, and counts in disjoint intervals become*
 234 *independent.*

235 *Proof sketch.* Layers use disjoint random streams and no cross-layer coupling, so the opening streams
 236 are independent point processes each converging to Poisson (Proposition 1); the claim is a special case
 237 of the superposition theorem of Grigelionis [10]. $\square$

238 **Proposition 3** (Finite- p dispersion correction). *For finite p , the asymptotic index of dispersion of a*
 239 *single layer’s opening counts equals the squared coefficient of variation of W :*

$$D_\infty = \text{CV}^2(W) = 1 - p\left(1 - \frac{1}{\kappa}\right). \quad (2)$$

240 *Proof sketch.* $E[W] = \mu/p$ and $\text{Var}[W] = \text{Var}(X)/p + E[X]^2(1-p)/p^2$ give (2) directly; the dispersion
 241 of renewal counts converges to CV^2 of the inter-arrival law [13]. $\square$

242 At the battery configuration ($p \approx 0.004$, $\kappa \in [1.8, 4]$ across layers), (2) predicts dispersion between
 243 0.9970 and 0.9982: a 0.18–0.30% deviation *below* unity (Gamma sojourns with $\kappa > 1$ are more regular
 244 than exponential), an order of magnitude smaller than the sampling noise of a dispersion estimate from
 245 200 windows ($\text{sd} \approx \sqrt{2/199} \approx 0.10$). The prediction is thus: *not exactly Poisson, provably Poisson in*
 246 *the limit, and statistically indistinguishable from Poisson at any realistic sample size.*

247 Two boundaries of the claim deserve emphasis. First, it is *conditional on one built world*. Across
 248 construction seeds the rate Λ itself varies, making unconditional counts *mixed* Poisson with $\text{Var}[N] =$
 249 $E[\Lambda]T + T^2 \text{Var}[\Lambda] > E[N]$ [15]: overdispersion across differently built models is expected and real
 250 (Section 7.3). Second, the propositions thin transitions i.i.d., which is exactly mechanism A. Mechanism B’s
 251 matching predicate depends on the *joint* cross-layer state, a dependence outside the assumptions;
 252 Sections 6.3 and 6.4 measure what it does.

253 5 Experimental design

254 5.1 Configurations

255 The *battery configuration* (mechanism A) assigns each unknown-bearing layer an unknown transition
 256 mass of 0.4% and simulates two million miles, yielding 39,227 episodes (one per 50.98 mi on average)—
 257 deliberately high statistical power. The *reference configuration* (mechanism B, version 6) is the one
 258 summarized in Tables 1 and 2: 100 planted combination rules, a 20,000 h horizon, and unknown episodes
 259 arriving three orders of magnitude more rarely (one per $\approx 2,000$ mi), which stress-tests the diagnostics in
 260 the regime industrial practice actually faces.

261 5.2 Statistical tests

262 We use three complementary tests. (T1) *Dispersion.* For counts $k_1, \dots, k_n$ in n disjoint windows,
 263 $(n-1)D \sim \chi^2_{n-1}$ under the Poisson hypothesis [13]; we report two-sided p -values. (T2) *Goodness of fit.*
 264 A χ^2 test of the empirical count histogram against $\text{Poisson}(\bar{k})$, bins merged to expected count ≥ 5 , one
 265 degree of freedom charged for the estimated mean. (T3) *Inter-arrival law.* A Kolmogorov–Smirnov test
 266 [16] of the observed inter-opening distances against the exponential distribution with matched mean;
 267 exponential gaps are equivalent to the Poisson property for a renewal stream.

5.3 The verification battery (mechanism A)

Three experiments target the three Poisson properties. (A) *Replication*: 300 independent runs of 3,000 mi over the *same built world* (construction seeds fixed; run-time seeds varied), so that under the hypothesis the 300 counts are i.i.d. Poisson; both T1 and T2 are applied. (B) *Windows*: counts in 200 disjoint 10,000 mi windows of the two-million-mile run (T1). (C) *Gaps*: the 39,227 inter-arrival distances of the same run (T3).

5.4 Diagnostics and manifestation merging (mechanism B)

For the reference run we apply T1–T3 to the 495 episode openings (99 complete 10,000 mi windows) and complement them with a mechanistic gap analysis: classifying every short inter-arrival by whether the *same* rule re-opened, and re-running all tests after merging same-rule re-openings within a mileage threshold into a single *manifestation*. The merge threshold is varied (10–100 mi) to check robustness.

5.5 Sensitivity study suite

Twenty-one one-factor studies sweep every model control—class percentage maps, combination budgets and size allocations, per-layer sojourn means and variances, class duration profiles, concentration, self-transition policy, horizon, speed, window size, minimum-duration clamp, sojourn distribution family, and whole-model seed variability—with at least ten levels per numeric control and three replicates per level: 1,160 simulations in total. A dedicated aggregation study normalizes each factor’s effect range by the seed-only standard deviation (a signal-to-noise ratio, SNR), so that factor effects are judged against genuine run-to-run variability. Seed-only variability is itself estimated from 20 whole-model reseeds.

6 Results

6.1 Model verification

Before testing distributional hypotheses we verify that the simulator realizes its configuration. Class transition masses are exact by construction and the empirical unknown selection share across the reference run is 10.02% against the 10.00% target (Figure 2); realized per-layer mean sojourns lie within $\approx 4\%$ of their configured blended values (e.g. street: 381 s configured, 366 s realized); and realized street-layer visit shares match configured probabilities to within a tenth of a percentage point. Mileage–time consistency holds exactly (10^6 mi / 50 mph = 20,000 h).

6.2 Mechanism A: the verification battery

All three experiments retain the Poisson hypothesis (Table 3, Figure 3). The replication experiment gives dispersion 1.078 across 300 runs ($p = 0.34$) with a count histogram compatible with Poisson (58.6) ($p = 0.14$); window counts of the two-million-mile run give dispersion 0.956 ($p = 0.68$); and the 39,227 inter-arrival distances are compatible with the exponential law (KS = 0.0055, $p = 0.18$). The point predicted by Proposition 3—a dispersion deficit of two to three parts per thousand—is far below the resolution of these tests, exactly as intended.

6.3 Mechanism B: the reference run

The reference run produced 495 unknown episodes in one million miles (Table 4): 481 from size-3 rules and 14 from size-4 rules, none from sizes 5 and 6, whose planted masses ($\sim 10^{-10}$) make them unobservable at this horizon—an internally consistent picture, since expected counts scale with rule mass. Forty-eight of the 100 rules fired at least once. Episodes are brief (mean 20.9 s, 0.29 mi) and unknown occupancy is 0.0144% of driving time; the mean inter-arrival distance is 2,018 mi. Episode accumulation over mileage is steady (Figure 4).

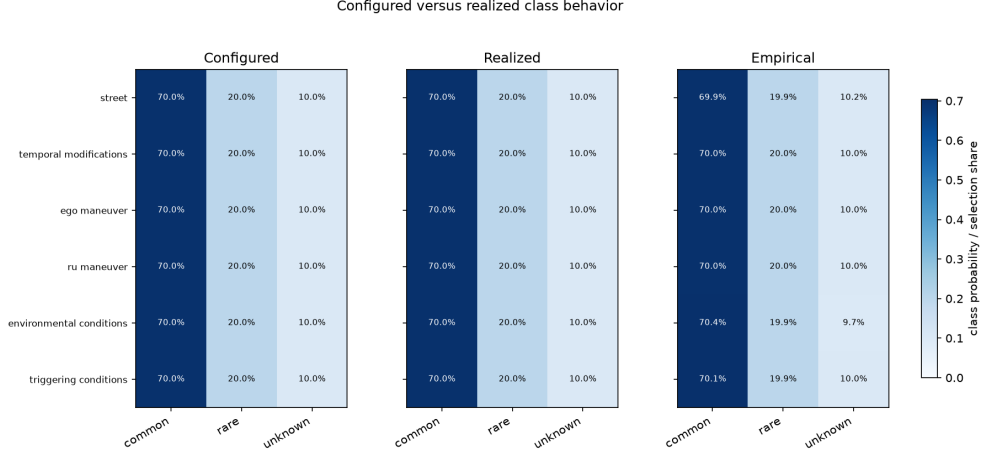


Figure 2: Configured versus realized class behaviour, per layer: configured selection percentages (left), realized permanent transition masses (centre; exact by construction), and empirical selection shares over the 4.08×10^6 -event reference run (right).

Table 3: Verification battery for mechanism A (element-route unknowns, $p \approx 0.004$ per transition). All tests are two-sided at the 5% level.

Experiment	Design	Statistic	p -value	Verdict
A: Replication (T1)	300 runs $\times$ 3,000 mi, same world	$D = 1.078$	0.34	retain
A: Replication (T2)	count histogram vs. Poisson(58.6)	$\chi^2 = 34.8$ (27 d.o.f.)	0.14	retain
B: Windows (T1)	200 windows $\times$ 10,000 mi	$D = 0.956$	0.68	retain
C: Gaps (T3)	39,227 inter-arrivals vs. exponential	KS = 0.0055	0.18	retain

Distributional diagnostics, however, are more interesting than for mechanism A (Table 5, first column). The marginal *shape* of window counts is fully compatible with Poisson (χ^2 GOF $p = 0.82$), but the variance runs high: $D = 1.355$ over the 99 windows (mean 4.93, variance 6.68; Figure 5), with a two-sided $p \approx 0.02$ —borderline but suggestive overdispersion. The inter-arrival test is unambiguous: the exponential law is rejected (KS = 0.112, $p < 0.001$), with squared coefficient of variation 1.20 and a pronounced excess of very short gaps: 37 gaps shorter than one mile against ≈ 0.3 expected under the exponential fit (Figure 6).

6.4 The clustering mechanism, and Poisson at the manifestation level

Classifying the short gaps identifies the cause. Of the 73 inter-arrivals shorter than 100 mi (about 24 expected under exponential gaps), 46 are re-openings of the *same rule* that had just closed. The mechanism is structural. A combination rule stays matchable only while its slow-layer rare elements persist: mean sojourns in the street, temporal-modification, and environment layers are minutes long (6, 12, and 37 min respectively, selection-weighted), while the fast manoeuvre layers turn over every 30–60 s. When a fast layer briefly leaves the matching state and returns—or a triggering element expires and is promptly re-drawn—the same persistent real-world condition is recorded as several distinct episodes in quick succession. This is a mild, mechanistically transparent residue of the same counting phenomenon that episode semantics eliminated for single-layer unknowns (Section 3.5): mechanism B’s matching predicate spans layers, so a single-layer flicker can interrupt it without ending the underlying condition.

The natural repair is definitional: count *manifestations*, merging re-openings of the same rule within a short mileage window into one event, the episode-level analogue of “the same FI condition, briefly interrupted”. Merging at a 10 mi threshold coalesces 45 of the 495 openings (9.1%) into predecessors. Every diagnostic then becomes fully compatible with the Poisson hypothesis (Table 5): dispersion drops

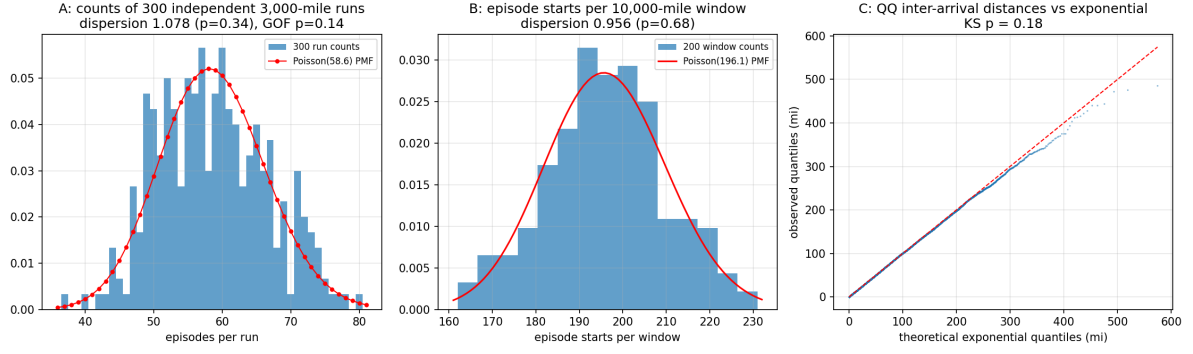


Figure 3: Mechanism A verification battery. Left: counts of 300 replicated 3,000 mi runs against the fitted Poisson probability mass function. Centre: counts per 10,000 mi window of the two-million-mile run against Poisson(196.1). Right: quantile–quantile plot of inter-arrival distances against the exponential distribution.

Table 4: Reference run (mechanism B, version 6): headline quantities.

Quantity	Value
Simulated time / mileage	20,000 h / 1,000,000 mi (at 50 mph)
Event-loop steps / wall-clock time	4,079,575 / 11.7 s
Unknown episodes	495 (C3: 481; C4: 14; C5, C6: 0; none truncated)
Rules fired / planted	48 / 100 (maximum 38 episodes for one rule)
Episode duration	mean 20.9 s (0.29 mi); median 16.4 s; p90 42.6 s; max 137 s
Unknown occupancy	10,362 s = 0.0144% of simulated time
Inter-arrival distance	mean 2,018 mi; median 1,368 mi
Episode rate	495 per million miles

to 1.084 ($p = 0.53$) and the exponential law is retained ($KS = 0.029$, $p = 0.84$). The conclusion is robust to the merge threshold: 50 and 100 mi thresholds give $D = 1.064$ ($p = 0.63$) and identical KS conclusions ($p = 0.89$), and a rule-agnostic debounce at 10 mi gives $D = 0.978$ ($p = 0.91$). The entire measured deviation is therefore re-opening book-keeping, not genuine clustering of distinct unknown conditions.

6.5 Sensitivity analysis

Figure 7 summarizes the 21-study suite. Seed-only variability is itself substantial—the whole-model reseeding standard deviation of the episode rate is $\approx 28\%$ of its mean—which the SNR normalization takes as the noise floor. Four factor groups dominate exposure: the element-class percentage map (SNR 123; a 3,400% rate range across its tested levels), the selection-class percentage map (SNR 46), the combination budget (SNR 19), and the rarity duration multipliers (SNR 9), followed by combination size allocation, per-layer mean sojourns, and per-layer element counts (SNR 5–6). Three controls show no measurable effect on the rate: average speed and mileage-window size, which merely relabel the time axis, and per-layer sojourn *variance*—a null predicted by Proposition 1, whose limiting rate $(m - s)/\mu$ depends on mean sojourns and masses but not on sojourn variability. We emphasize that the tested grids are deliberately wide screening ranges; the ranking orders the tested ranges rather than any unit-free notion of importance.

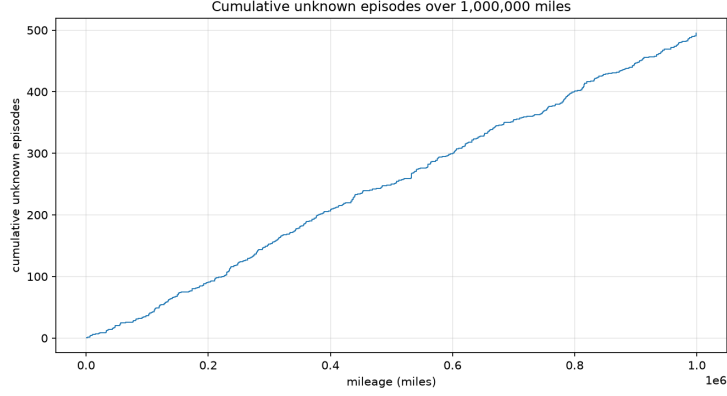


Figure 4: Cumulative unknown episodes over the one-million-mile reference run.

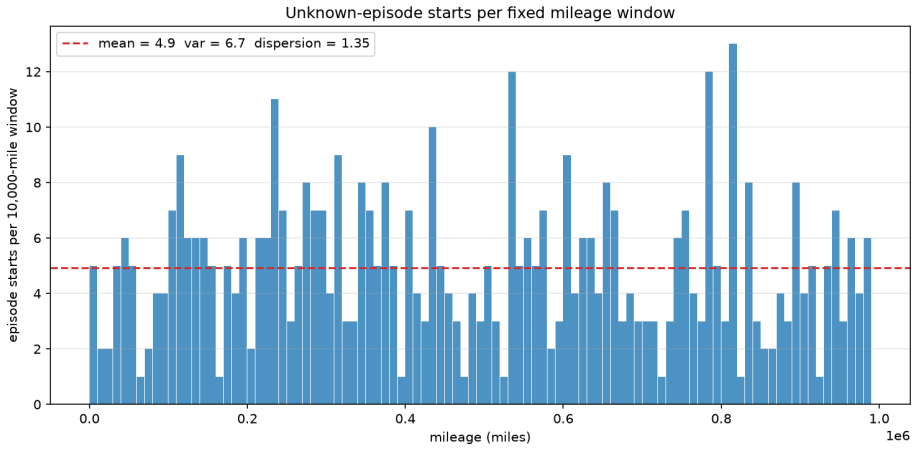


Figure 5: Mechanism B reference run: unknown-episode starts per 10,000 mi window (99 complete windows; mean 4.93, variance 6.68, dispersion 1.355).

7 Discussion

7.1 Support for the Poisson assumption

Within a fixed operational world, the assumption comes out of this study well—and not vacuously, since nothing in the generative model imposes it. For element-route unknowns the result is airtight: theory predicts Poisson in the rare-event limit with a quantified, negligible finite- p correction, and a high-power battery (39,227 events; 300 replicated runs) retains every Poisson property. For combination-type unknowns—the mechanism closest to how mature programmes actually encounter unknowns—the same conclusion holds at the level of *distinct manifestations*, after a definitional correction that is itself mechanistically motivated. The miles between manifestations are memoryless; window counts superpose; the arithmetic that converts observed event-free mileage into rate bounds [1] is sound under the conditions simulated here.

7.2 Counting matters: episodes, encounters, manifestations

The study’s sharpest practical lesson is that the Poisson question is ill-posed until the *unit of counting* is fixed. Three conventions appeared: encounters (every new tuple; $D \approx 12$), episodes (maximal matched periods; $D = 1.355$ for mechanism B), and manifestations (debounced episodes; $D = 1.08$, fully Poisson). The graduation is monotone: each step merges records that correspond to one underlying condition. Field-data programmes face exactly the same choice when an FI trigger flickers—e.g. a

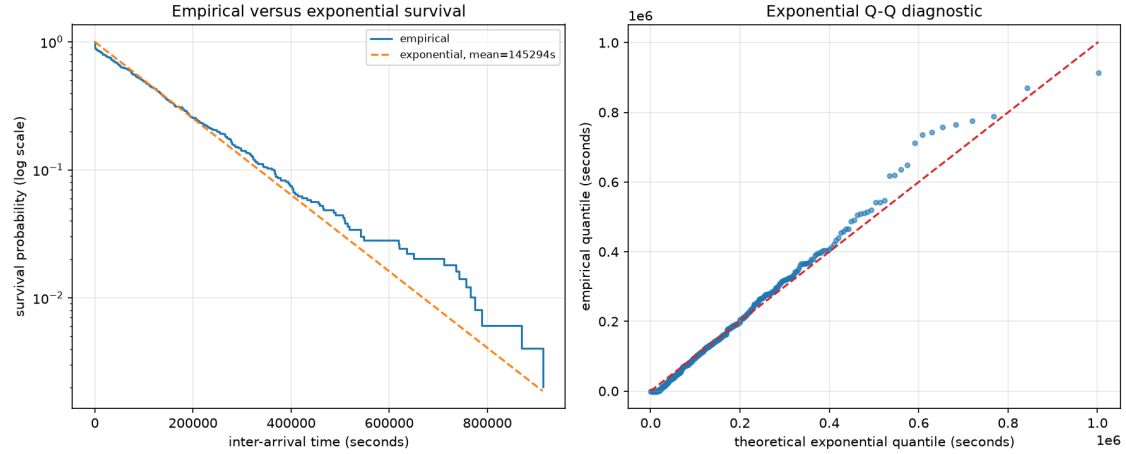


Figure 6: Mechanism B reference run: inter-arrival diagnostics. Left: empirical survival function of inter-arrival times against the fitted exponential (log scale). Right: exponential quantile–quantile plot. The excess of very short gaps and the heavy upper tail both reflect episode clustering (Section 6.4).

Table 5: Mechanism B diagnostics before and after manifestation merging (same-rule re-openings within 10 mi counted as one manifestation). Window tests use 10,000 mi windows; p -values two-sided.

	Raw episodes	Merged manifestations
Events	495	450
Dispersion index D (T1)	1.355	1.084
two-sided p	0.022	0.53
Count GOF vs. Poisson (T2), p	0.82	—
KS vs. exponential gaps (T3)	0.112	0.029
p	< 0.001	0.84
Gaps < 1 mi (expected under exp.)	37 (≈ 0.3)	—

perception insufficiency that recurs within the same rain shower—and our results quantify how strongly the choice biases dispersion estimates. For validation-mileage arithmetic, counts should be taken at the manifestation level; where raw episode logs are used, the measured dispersion index should be carried into the budget as a multiplicative safety factor rather than assumed to be one.

7.3 The boundary: model-level uncertainty

The Poisson claim is conditional on one built world. Across construction seeds—worlds that are statistically exchangeable under the same configuration—the episode rate varies with a standard deviation near 28% of its mean, making unconditional counts mixed-Poisson and hence overdispersed [15]. The practical reading is sobering: even if occurrences are perfectly Poisson given the world, uncertainty about the world itself (catalogue composition, rarity masses, exposure profile) dominates the uncertainty of any single-rate extrapolation. A validation-mileage figure derived from one Poisson rate estimate therefore understates total uncertainty; budgets should either condition on a well-characterized operational design domain or inflate targets for the between-world variance component. The same caution applies to decomposing acceptance criteria: additivity of rates survives superposition, but the implicit assumption that each class’s rate is a known constant does not.

7.4 Implications for practice

Three concrete recommendations follow. First, define the counted event precisely—one manifestation per underlying condition—and document the debounce rule; the choice moves dispersion by an order of

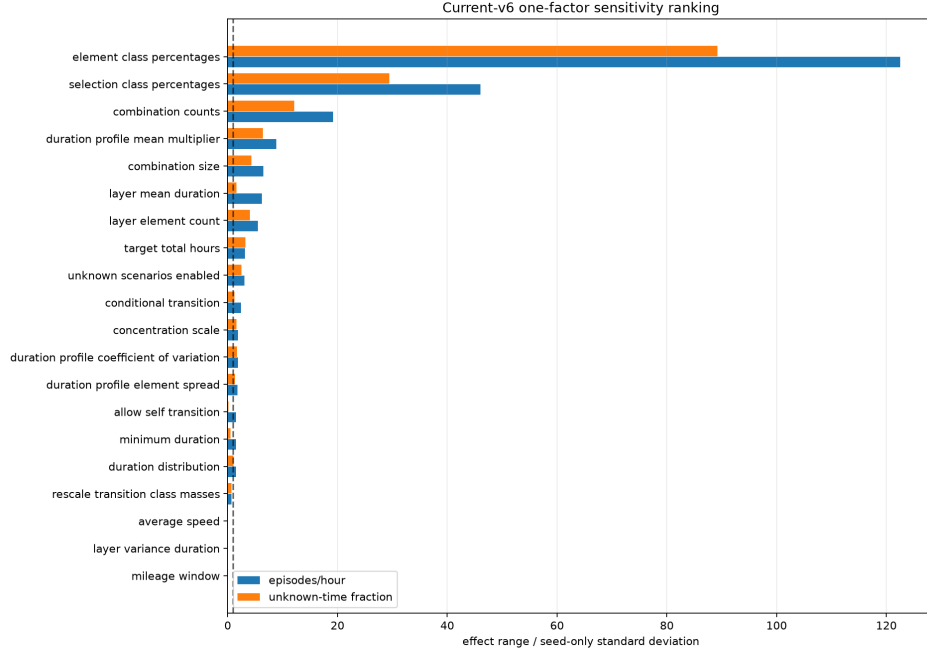


Figure 7: One-factor sensitivity ranking across the 21-study suite (1,160 simulations): effect range of each factor divided by the seed-only standard deviation (SNR), for the episode rate and for unknown-time occupancy.

magnitude. Second, monitor dispersion, not only rates: window-count dispersion is cheap to compute from any event log, has a known null distribution, and directly tests the assumption that mileage-based confidence arithmetic relies on. Third, treat single-number rate extrapolations as conditional statements; where the operational domain is heterogeneous or the exposure model uncertain, use the measured dispersion index (or an explicit mixed-Poisson model) to widen validation targets.

7.5 Limitations

The framework is deliberately abstract. Element catalogues, rarity splits, duration multipliers, and combination budgets are design defaults, not field calibrations; consequently, absolute rates (495 episodes per million miles in the reference run) carry no field meaning—only distributional shapes and their mechanisms do. Layers evolve independently in the studied configurations; real driving couples them (weather changes behaviour), and although the engine supports conditional cross-layer transitions, a systematic study of dependence-induced dispersion remains future work. The manifestation-merging analysis of mechanism B rests on one reference world at one horizon; multi-seed replication of the merged-count dispersion, analogous to battery experiment A, would tighten it. Finally, the mapping from “unknown scenario occurrence” to “FI manifestation” is programme-specific: we model the former and inherit the latter’s statistics only insofar as the triggering relation is one-to-one.

8 Conclusion

We asked whether the occurrence of unknown scenarios follows a Poisson distribution—the assumption beneath validation-mileage calculations and acceptance-criteria decompositions for automated driving systems—and answered it with a generative test: a reproducible six-layer semi-Markov simulator in which nothing Poisson is built in, only the layered structure of driving and controlled rarity. Conditional on a fixed world, the answer is affirmative and now quantified. Rare per-transition unknowns are provably Poisson in the limit, with a finite-probability deviation of a few parts per thousand, and pass a high-power empirical battery. Combination-type unknowns—individually known rare elements in never-anticipated

constellations—are Poisson at the level of distinct manifestations; their raw episode logs exhibit a 35% dispersion excess caused entirely by same-rule re-openings inside persistent slow-layer states, a counting artefact with a one-line remedy. Across differently built worlds, by contrast, counts are mixed-Poisson and overdispersed, and the between-world component ($\approx 28\%$ rate variability under reseeding) should be budgeted explicitly.

Future work will calibrate layer catalogues and sojourn scales to field data, extend the dispersion analysis to conditional (coupled) transition models, replicate the manifestation-level result across many built worlds, and study interaction effects beyond the one-factor screening reported here.

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Data and code availability

The simulator (Python), configuration files, automated test suite (39 tests), run outputs (episode and window logs, summary statistics), and the complete 21-study sensitivity suite will be made available [repository/DOI to be completed; available from the authors on reasonable request in the interim].

References

- [1] Nidhi Kalra and Susan M. Paddock. Driving to safety: How many miles of driving would it take to demonstrate autonomous vehicle reliability? *Transportation Research Part A: Policy and Practice*, 94:182–193, 2016. doi: 10.1016/j.tra.2016.09.010.
- [2] Walther Wachenfeld and Hermann Winner. The release of autonomous vehicles. In Markus Maurer, J. Christian Gerdes, Barbara Lenz, and Hermann Winner, editors, *Autonomous Driving: Technical, Legal and Social Aspects*, pages 425–449. Springer, Berlin, Heidelberg, 2016. doi: 10.1007/978-3-662-48847-8_21.
- [3] Stefan Riedmaier, Thomas Ponn, Dieter Ludwig, Bernhard Schick, and Frank Diermeyer. Survey on scenario-based safety assessment of automated vehicles. *IEEE Access*, 8:87456–87477, 2020. doi: 10.1109/ACCESS.2020.2993730.
- [4] Till Menzel, Gerrit Bagschik, and Markus Maurer. Scenarios for development, test and validation of automated vehicles. In *2018 IEEE Intelligent Vehicles Symposium (IV)*, pages 1821–1827, 2018. doi: 10.1109/IVS.2018.8500406.
- [5] International Organization for Standardization. ISO 26262:2018 road vehicles — functional safety. Technical report, ISO, Geneva, Switzerland, 2018.
- [6] International Organization for Standardization. ISO 21448:2022 road vehicles — safety of the intended functionality. Technical report, ISO, Geneva, Switzerland, 2022.
- [7] Philip Koopman and Michael Wagner. Challenges in autonomous vehicle testing and validation. *SAE International Journal of Transportation Safety*, 4(1):15–24, 2016. doi: 10.4271/2016-01-0128.
- [8] Alfréd Rényi. A characterization of Poisson processes. *Magyar Tudományos Akadémia Matematikai Kutató Intézetének Közleményei*, 1:519–527, 1956. In Hungarian.
- [9] David R. Cox. *Renewal Theory*. Methuen, London, 1962.

- 444 [10] Bronius Grigelionis. On the convergence of sums of random step processes to a Poisson process.
445 *Theory of Probability & Its Applications*, 8(2):177–182, 1963. doi: 10.1137/1108017.
- 446 [11] PEGASUS Project Consortium. PEGASUS method: An overview. <https://www.pegasusprojekt.de>, 2019. German federal research project on scenario-based validation of
447 highly automated driving.
448
- 449 [12] Maïke Scholtes, Lukas Westhofen, Lara Ruth Turner, Katrin Lotto, Michael Schuldes, Hendrik
450 Weber, Nicolas Wagener, Christian Neurohr, Martin Herbert Bollmann, Franziska Körtke, Johannes
451 Hiller, Michael Hoss, Julian Bock, and Lutz Eckstein. 6-layer model for a structured description
452 and categorization of urban traffic and environment. *IEEE Access*, 9:59131–59147, 2021. doi:
453 10.1109/ACCESS.2021.3072739.
- 454 [13] David R. Cox and Peter A. W. Lewis. *The Statistical Analysis of Series of Events*. Methuen,
455 London, 1966.
- 456 [14] Richard Arratia, Larry Goldstein, and Louis Gordon. Two moments suffice for Poisson ap-
457 proximations: the Chen–Stein method. *The Annals of Probability*, 17(1):9–25, 1989. doi:
458 10.1214/aop/1176991491.
- 459 [15] Jan Grandell. *Mixed Poisson Processes*. Chapman & Hall, London, 1997.
- 460 [16] Frank J. Massey. The Kolmogorov–Smirnov test for goodness of fit. *Journal of the American*
461 *Statistical Association*, 46(253):68–78, 1951. doi: 10.1080/01621459.1951.10500769.